

Task 4: Feature Encoding & Scaling - Interview Questions & Answers

1. Difference between Label Encoding and One-Hot Encoding?

Label Encoding: Converts categories into numerical labels. Example: Male = 1, Female = 0.

One-Hot Encoding: Creates separate binary columns for each category. Example: City_Hyderabad, City_Chennai.

Label Encoding is mainly used for ordinal or binary data, while One-Hot Encoding is used for nominal data.

2. When should you use Standardization vs Normalization?

Standardization: Converts data to mean = 0 and standard deviation = 1. It is useful for algorithms like KNN, Logistic Regression, and SVM.

Normalization: Scales values between 0 and 1. It is useful for Neural Networks and Deep Learning models.

3. What is the Dummy Variable Trap?

Dummy Variable Trap occurs when one encoded variable can be predicted from another variable, causing multicollinearity problems in machine learning models.

It can be avoided using:

```
pd.get_dummies(drop_first=True)
```

4. Why do distance-based algorithms like KNN require scaling?

Distance-based algorithms calculate distance between data points. If one feature has very large values, it dominates the distance calculation.

Scaling ensures all features contribute equally to the model.

5. Can we apply One-Hot Encoding to high-cardinality features?

Yes, but it may create too many columns and increase memory usage.

Problems include:

- High dimensionality
- Increased computation time
- Overfitting risk

Alternative methods include Target Encoding and Frequency Encoding.